Operations Research & Game Theory: LP-Based Decision Optimization using R

End-to-end Operations Research project solving three real-world optimization problems — production planning, multi-product blending, and zero-sum gameß theory — using Linear Programming with R (`lpSolve`, `lpSolveAPI`). Includes graphical solutions, sensitivity analysis, and mixed-strategy Nash equilibrium derivation.

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Q1. Garment Factory – Linear Programming Model and Graphical Solution

Problem Overview

A garment manufacturer produces two products—shirts (X) and pants (Y)—using three limited resources: cutting, sewing, and packaging labor. Each worker provides 8 effective hours (480 minutes) per day, and product-specific time requirements for each operation are provided. Profits are $10 per shirt and $8 per pant. Daily shirt demand is limited to 180 units.

	Time (Minutes) Per Operation			
	Cutting	**Sewing**	**Packaging**	**Profit per unit ($)**
Shirts(X)	40	40	20	10
Pants(Y)	20	100	20	8

The objective is to determine the **production mix that maximizes daily profit**, subject to labor and demand constraints.

A) Why Linear Programming for the Garment factory?

A **linear programming (LP)** model is well suited to the garment factory case because both the objective and the operational constraints can be represented using linear relationships.

- The primary objective of the factory is to **maximise daily profit**. This profit is directly determined by the number of shirts and pants produced, with each unit contributing a fixed and known profit. As a result, the objective function is inherently linear in nature.

- **Resource limitations:** The factory operates under resource limitations. There are a fixed number of workers in the cutting (20), sewing (50), and packaging (14) departments, and a limited number of productive hours (8 hours) per day. Each product requires a fixed amount of processing time per unit in each department, and the total available working time is limited by the number of workers and productive hours. Consequently, the total time consumed in cutting, sewing, and packaging can be expressed as a linear combination of the quantities of shirts and pants produced.

- Moreover, there is a direct proportional relationship between inputs and outputs. For instance, producing an additional unit of a garment increases both the required processing time and the profit by a constant amount. This proportionality between resource usage and production volume satisfies a fundamental assumption of linear programming, namely that both costs and returns scale linearly with output.

- The given problem requires determining specific numerical values - the number of shirts and pants to produce daily - which serve as the decision variables in a Linear Programming model.

Additionally, all d**ecision variables are non-negative**, and there is no uncertainty or non-linear interaction between variables. Since the problem involves allocating limited resources to competing activities in order to optimize a single objective, it fits perfectly within the framework of linear programming.

B) Linear Programming Formulation

*To formulate the linear programming (LP) model, we first define the **decision variables.** Let -*

X - Number of **shirts** produced per day

Y - Number of **pants** produced per day

Objective Function

The objective of the factory is to **maximize total daily profit** with Profit per shirt = 10$ and Profit per pant = 8$.

$$MaxZ = 10X + 8Y$$

Constraints Formulation

1. Cutting

→ Number of cutting workers = 20
→ Working time per worker = 8 hours = 480 minutes
→ Total cutting time available per day = 20 × 480 = 9600 minutes

Time required per unit:

→ Shirt: 40 minutes; Pant: 20 minutes

$$40X + 20Y \leq 9600$$

2. Sewing

→ Number of sewing workers = 50

→ Total sewing time available per day = 50 × 480 = 24000 minutes

Time required per unit:

→ Shirt: 40 minutes; Pant: 100 minutes

$$40X + 100Y \leq 24000$$

3. Packaging

→ Number of packaging workers = 14
→ Working time per worker = 8 hours = 480 minutes

→ Total packaging time available per day = 14 × 480 = 6720 minutes

Time required per unit:

→ Shirt: 20 minutes; Pant: 20 minutes

$$20X + 20Y \leq 6720$$

4. Demand Constraint for Shirts

→ Maximum daily demand for shirts = 180 units

$$X \leq 180$$

5. Non Negative

$$X,\ Y \geq 0$$

LP Model Summary

$$MaxZ = 10X + 8Y$$

Subject to :

$$40X + 20Y \leq 9600 \quad \text{(Cutting)}$$

$$40X + 100Y \leq 24000 \quad \text{(Sewing)}$$

$$20X + 20Y \leq 6720 \quad \text{(Packaging)}$$

$$X \leq 180 \quad \text{(Demand Limit)}$$

$$X, Y \geq 0 \quad \text{(Non-negativity)}$$

C) Graphical Solution to find the optimal points

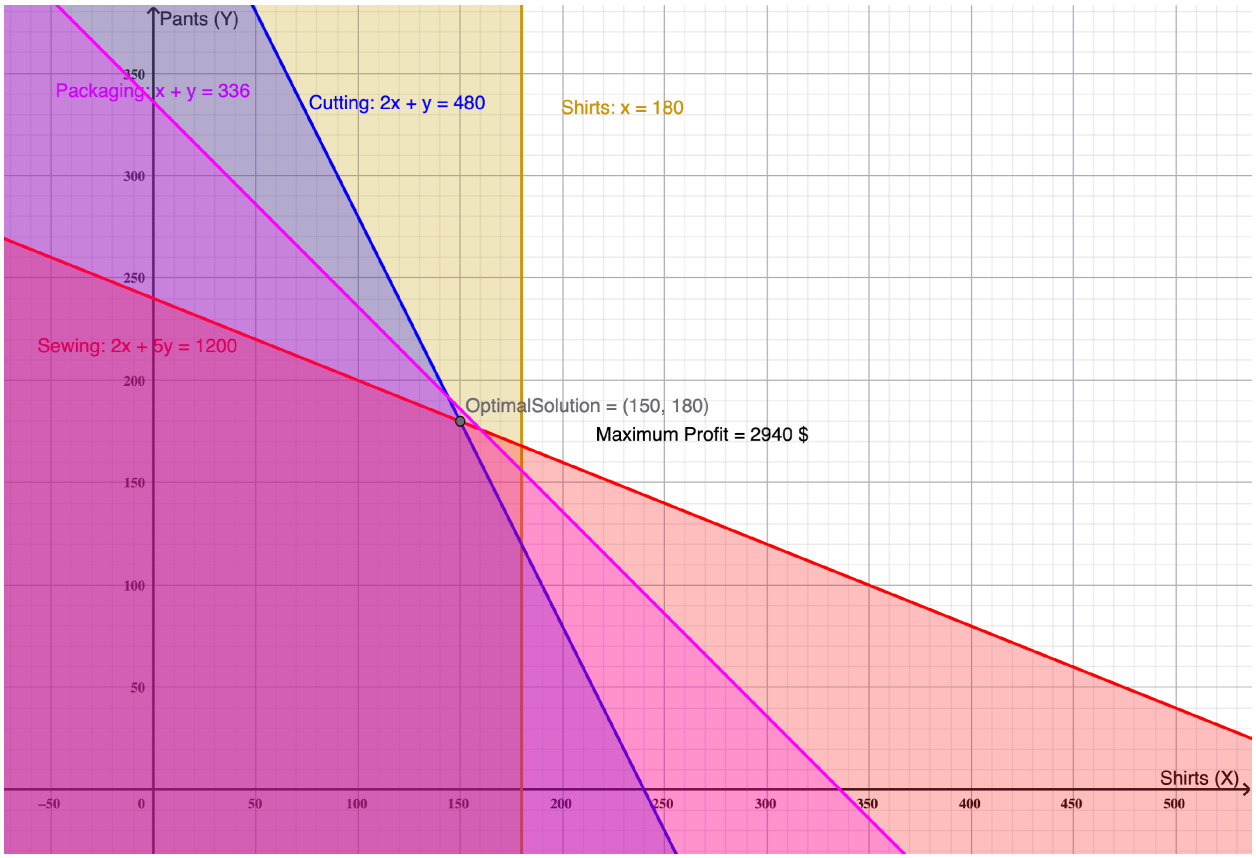

Fig 1: The graphical representation of all constraints for Garment factory case

The feasible region is the intersection of all constraint half-spaces.

The optimal solution lies at the intersection of the cutting and sewing constraints.

Optimal Production Plan

Shirts: $X^* = 150$; Pants: $Y^* = 180$

Maximum Profit: $Z^* = 10\ (150) + 8\ (180) = \$\ 2940$

Binding Constraints

- *Cutting capacity*

- *Sewing capacity*

*Packaging capacity is **not binding** (120 minutes unused).*

D) Sensitivity Analysis for Profit per Shirt

The optimal solution occurs at the intersection of the Cutting and Sewing constraints. For this corner point to remain at the maximum, the slope of the objective function line must stay between the slopes of these two binding constraint lines.

Determine Slopes of Binding Constraints

Cutting Constraint

$$40X + 20Y = 9600$$
$$Y = -2X + 480$$

→ Slope ($m_{cutting}$) = **-2**

Sewing Constraint

$$40X + 100Y = 24000$$
$$Y = -0.4X + 2400$$

→ Slope (m_{sewing}) = **-0.4**

Determine Slope of the Objective Function

Let C_s = profit per shirt (originally $10). And the profit per pant remain fixed $8.

$$Z = CsX + 8Y$$

$$Y = -\frac{C_s}{8}X + \frac{Z}{8}$$

→ Slope (m_{obj}) = $-\frac{C_s}{8}$

Calculating the Range

For the current solution to remain optimal, the slope of the profit line must lie between the slopes of the Cutting and Sewing lines:

$$M_{cutting} \leq m_{obj} \leq m_{sewing}$$

$$-2 \leq -\frac{C_s}{8} \leq -0.4 \qquad \text{[multiply by –8 to solve for } C_s\text{]}$$

$$3.2 \leq C_s \leq 16$$

The profit per shirt can range between **$ 3.20** and **$ 16.00** without changing the optimal production schedule of 150 shirts and 180 pants. If the profit drops below 3.20 $, the optimal mix will shift along the Sewing line, if it exceeds 16.00 $, the mix will shift along the Cutting line.

Q2. Optimization of Multi-Product Production Blending Using Linear Programming

Problem Overview

The factory produces three products - **Spring, Autumn, and Winter** - using three raw materials: **Cotton, Wool, and Silk**. Each product has a specified selling price and production cost, while each raw material has an associated purchase cost per ton. Additionally, each product must satisfy minimum composition requirements for Cotton, Wool, and Silk, and is subject to a maximum market demand.

	Sales Price	Production cost
Spring	$ 62	$ 7
Autumn	$ 47	$ 4
Winter	$ 71	$ 12

	Purchase Price
Cotton	$ 40
Wool	$ 46
Silk	$ 52

	Demand	min cotton Proportion	min wool proportion	min silk proportion
Spring	3500	51%	10%	2%
Autumn	4400	40%	20%	3%
Winter	4800	38%	30%	4%

The management must decide how much of each raw material to allocate to each product in order to maximise overall profit. This decision-making process is constrained by product demand limits, minimum material proportion requirements, and non-negativity conditions. Since there is no penalty for unmet demand, the challenge lies in optimally blending raw materials across products while respecting all constraints.

Given the linear nature of costs, revenues, and constraints, this problem can be effectively modelled and solved using linear programming techniques.

The objective of this problem is to **maximise the total profit** of the factory by determining the optimal quantities of Cotton, Wool, and Silk used in the production of Spring, Autumn, and Winter products, while satisfying demand limitations and minimum material composition requirements for each product.

A) Formulation of the Linear Programming (LP) model for the product blending problem

Decision Variables

Let X_{ij} be the amount (in tons) of Material i used to produce Product j .

- Indices for Materials (i): C (Cotton), W (Wool), S (Silk).
- Indices for Products (j): 1 (Spring), 2 (Autumn), 3 (Winter).

There are 9 decision variables in total:

- X_{C1}, X_{W1}, X_{S1} (Materials for Spring)
- X_{C2}, X_{W2}, X_{S2} (Materials for Autumn)

- X_{C3}, X_{W3}, X_{S3} (Materials for Winter)

Parameters & Profit Calculation

We calculate the profit coefficient for each variable by subtracting the costs from the sales price.

Profit (C_{ij}) = Sales Price (j) - Production Cost (j) - Material Cost (i)

Product 1: Spring (Sales: $ 62 , Prod Cost: $7) Net Revenue: 62 – 7 = $55

- Cotton (X_{C1}): 55 - 40 = 15$
- Wool (X_{W1}): 55 - 46 = 9$
- Silk (X_{S1}): 55 - 52 = 3$

Product 2: Autumn (Sales: $47, Prod Cost: $4) Net Revenue: 47 – 4 = $43

- Cotton (X_{C2}): 43 - 40 = 3$
- Wool (X_{W2}): 43 - 46 = -3$ (Loss)
- Silk (X_{S2}): 43 - 52 = -9$ (Loss)

Product 3: Winter (Sales: $71, Prod Cost: $12) Net Revenue: 71 – 12 = $59

- Cotton (X_{C3}): 59 - 40 = 19$
- Wool (X_{W3}): 59 - 46 = 13$
- Silk (X_{S3}): 59 - 52 = 7$

Objective Function

Maximize total profit:

$$Max\,Z = 15X_{C1} + 9X_{W1} + 3X_{S1} + 3X_{C2} - 3X_{W2} - 9X_{S2} + 19X_{C3} + 13X_{W3} + 7X_{S3}$$

Constraints

Demand Constraints

The total weight produced for each product ($\sum_i X_{ij}$) must not exceed the maximum demand.

- Spring: $X_{C1} + X_{W1} + X_{S1} \leq 3500$
- Autumn: $X_{C2} + X_{W2} + X_{S2} \leq 4400$
- Winter: $X_{C3} + X_{W3} + X_{S3} \leq 4800$

Blending (Proportion) Constraints

The amount of a specific material must be at least a certain percentage of the total product weight. Formula: $X_{ij} \geq P\% \times \left(X_{Cj} + X_{Wj} + X_{Sj}\right)$

For Spring ($j = 1$):

$Cotton \geq 51\%$: $X_{C1} \geq 0.51\left(X_{C1} + X_{W1} + X_{S1}\right)$ => $0.49X_{C1} - 0.51X_{W1} - 0.51X_{S1} \geq 0$

$Wool \geq 10\,\%$: $X_{W1} \geq 0.10\left(X_{C1} + X_{W1} + X_{S1}\right)$ => $-0.10X_{C1} + 0.90X_{W1} - 0.10X_{S1} \geq 0$

$Silk \geq 2\,\%$: $X_{S1} \geq 0.02\left(X_{C1} + X_{W1} + X_{S1}\right)$ => $-0.02X_{C1} - 0.02X_{W1} + 0.98X_{S1} \geq 0$

For Autumn ($j = 2$):

$Cotton \geq 40\,\%$: $X_{C2} \geq 0.40\left(X_{C2} + X_{W2} + X_{S2}\right)$ => $0.60X_{C2} - 0.40X_{W2} - 0.40X_{S2} \geq 0$

$Wool \geq 20\ \%$: $X_{W2} \geq 0.20\left(X_{C2} + X_{W2} + X_{S2}\right)$ =>
$-\ 0.20X_{C2} + 0.80X_{W2} - 0.20X_{S2} \geq 0$

$Silk \geq 3\ \%$: $X_{S2} \geq 0.03\left(X_{C2} + X_{W2} + X_{S2}\right)$ => $-\ 0.03X_{C2} - 0.03X_{W2} + 0.97X_{S2} \geq 0$

For Winter ($j = 3$):

$Cotton \geq 38\ \%$: $X_{C3} \geq 0.38\left(X_{C3} + X_{W3} + X_{S3}\right)$ => $0.62X_{C3} - 0.38X_{W3} - 0.38X_{S3} \geq 0$

$Wool \geq 30\ \%$: $X_{W3} \geq 0.30\left(X_{C3} + X_{W3} + X_{S3}\right)$ =>
$-\ 0.30X_{C3} + 0.70X_{W3} - 0.30X_{S3} \geq 0$

$Silk \geq 4\ \%$: $X_{S3} \geq 0.04\left(X_{C3} + X_{W3} + X_{S3}\right)$ => $-\ 0.04X_{C3} - 0.04X_{W3} + 0.96X_{S3} \geq 0$

Non-Negativity

$$X_{ij} \geq 0\ ; \qquad \text{for all } i,\ j$$

Result and Interpretation

The results indicate that the **maximum achievable profit is $136,152**, as obtained through the implementation of the linear programming model using R. The optimal production plan recommends meeting the entire market demand for all three product lines, with production levels of **3,500 tons for Spring**, **4,400 tons for Autumn**, and **4,800 tons for Winter**.

Q3. Optimal Strategy Determination in a Sequential Two-Player Zero-Sum Game Using Linear Programming

Problem Overview

Two players participate in a strategic parlor game in which each player begins with three distinct chips: red, white, and blue. Each chip may be used only once, and the game is played over three successive rounds. In each round, both players simultaneously select one of their remaining chips, conceal their choice, and then reveal it to determine the payoff based on predefined rules. If both players select the same color, the round results in a draw with zero payoff; otherwise, the payoff is awarded to the player whose chip dominates according to the given payoff structure.

Winning chip	Payoff ($)
Red beats white	80
White beats blue	50
Blue beats red	30
Matching colors	0

Because each chip can be used only once, a player's overall strategy must specify the order in which the three chips are played across the three rounds. This creates a finite set of possible pure strategies for each player. The competitive and zero-sum nature of the game requires both players to adopt strategies that maximise their expected payoff while accounting for the possible actions of the opponent.

Given the complexity of the strategy space and the sequential payoffs, the game can be analysed using game theory and formulated as a linear programming problem to identify optimal mixed strategies, equilibrium payoffs, and the existence or absence of saddle points.

The objective of this problem is to formulate and solve a two-player zero-sum game using linear programming in order to:

- determine the optimal strategies for both players,
- compute the expected value (payoff) of the game, and
- assess whether the optimal strategies and equilibrium solutions are unique.

A) Payoff Matrix and Saddle Points

To formulate the payoff matrix, we must first define the possible strategies for both players based on the hints provided. A strategy consists of the order in which the three chips (Red, White, Blue) are played.

Defining Strategies

In this parlor game, each player has three chips: Red (R), White (W), and Blue (B), which are used sequentially. A strategy is defined by the specific order in which these chips are played. There are 3! = 6 possible permutations for each player: Let's denote the strategies by the order of chips played:

I. **RWB (Red > White > Blue)**
II. **RBW (Red > Blue > White)**

III. **WRB (White > Red > Blue)**
IV. **WBR (White > Blue > Red)**
V. **BRW (Blue > Red > White)**
VI. **BWR (Blue, White, Red)**

Construction of the Payoff Matrix

The payoff for each round is determined by the following rules:

- Red beats White (80) / White loses to Red (-80)
- White beats Blue (50) / Blue loses to White (-50)
- Blue beats Red (30) / Red loses to Blue (-30)
- Matching colors result in a draw (0).

*The total payoff for a pair of strategies is the sum of the payoffs from the three individual rounds. For instance, if Player I plays **RWB** and Player II plays **WBR**:*

Example Calculation (Strategy "RWB" vs "WBR"):

- Round 1: Player I (Red) vs Player II (White). Red wins. Payoff: +80
- Round 2: Player I (White) vs Player II (Blue). White wins. Payoff: +50
- Round 3: Player I (Blue) vs Player II (Red). Blue wins. Payoff: +30
- Total Payoff: 80 + 50 + 30 = 160

Example Calculation (Strategy "RWB" vs "BRW"):

- Round 1: Player I (Red) vs Player II (Blue). Red loses. Payoff: -30
- Round 2: Player I (White) vs Player II (Red). White loses. Payoff: -80
- Round 3: Player I (Blue) vs Player II (White). Blue loses. Payoff: -50
- Total Payoff: $-30 - 80 - 50 = -160

The Payoff Matrix

Following this logic for all 36 combinations, we obtain the payoff matrix

Player II → \ Player I ↓	RWB	RBW	WRB	WBR	BRW	BWR
RWB	0	0	0	160	-160	0
RBW	0	0	160	0	0	-160

WRB	0	-160	0	0	0	160
WBR	-160	0	0	0	160	0
BRW	160	0	0	-160	0	0
BWR	0	160	-160	0	0	0

Identifying the Saddle Points

To find the saddle point, we evaluate the security level for each player to determine if a stable solution exists in pure strategies. A saddle point exists if the Maximin value equals the Minimax value.

- Maximin (Player I): This is the maximum of the minimum gains Player I can secure for each strategy.
  - All row minimums are {-160, -160, -160, -160, -160, -160}
  - Maximin = -160
- Minimax (Player II): This is the minimum of the maximum losses Player II would face
  - All column maximums are {160, 160, 160, 160, 160, 160}
  - Minimax = 160

Since **Maximin (-160)** $\neq$ **Minimax (160)**, there are no saddle points in this game. This indicates that a pure strategy will not result in an equilibrium, and both players must instead resort to mixed strategies to maximize their security level.

B) Linear Programming Models for Player I and Player II

To solve for the optimal mixed strategies in this zero-sum game, we formulate two linear programming (LP) models—one for each player.

Decision Variables

Let $S = \{S_1, S_2, S_3, S_4, S_5, S_6\}$ be the set of strategies.
where: S_1 = RWB; S_2 = RBW, S_3 = WRB, S_4 = WBR, S_5 = BRW, S_6 = BWR.

For Player I: Let x_i be the probability that Player I chooses strategy S_i for i = 1, . . . , 6.
For Player II: Let y_j be the probability that Player II chooses strategy S_j for j = 1, . . . , 6.
Let v represent the value of the game.

Player I's Linear Programming Model

Player I aims to **maximize** their minimum guaranteed payoff (v), regardless of the strategy chosen by Player II. Based on the columns of the payoff matrix derived above, the model is:

Maximize: v **Subject to:**

- Against P2 playing RWB: $0x_1 + 0x_2 + 0x_3 - 160x_4 + 160x_5 + 0x_6 \geq v$
- Against P2 playing RBW: $0x_1 + 0x_2 - 160x_3 + 0x_4 + 0x_5 + 160x_6 \geq v$
- Against P2 playing WRB: $0x_1 + 160x_2 + 0x_3 + 0x_4 + 0x_5 - 160x_6 \geq v$

- Against P2 playing WBR: $160x_1 + 0x_2 + 0x_3 + 0x_4 - 160x_5 + 0x_6 \geq v$
- Against P2 playing BRW: $-160x_1 + 0x_2 + 0x_3 + 160x_4 + 0x_5 + 0x_6 \geq v$
- Against P2 playing BWR: $0x_1 - 160x_2 + 160x_3 + 0x_4 + 0x_5 + 0x_6 \geq v$
- Probability Sum: $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 1$
- Non-negativity: $x_i \geq 0$ for all $i = 1, \ldots, 6$; v is unrestricted.

Player II's Linear Programming Model

Player II aims to minimize the maximum payoff they must pay to Player I. Based on the rows of the payoff matrix, the model is:

Minimize: v Subject to:

- Against P1 playing RWB: $0y_1 + 0y_2 + 0y_3 + 160y_4 - 160y_5 + 0y_6 \leq v$
- Against P1 playing RBW: $0y_1 + 0y_2 + 160y_3 + 0y_4 + 0y_5 - 160y_6 \leq v$
- Against P1 playing WRB: $0y_1 - 160y_2 + 0y_3 + 0y_4 + 0y_5 + 160y_6 \leq v$
- Against P1 playing WBR: $-160y_1 + 0y_2 + 0y_3 + 0y_4 + 160y_5 + 0y_6 \leq v$
- Against P1 playing BRW: $160y_1 + 0y_2 + 0y_3 - 160y_4 + 0y_5 + 0y_6 \leq v$
- Against P1 playing BWR: $0y_1 + 160y_2 - 160y_3 + 0y_4 + 0y_5 + 0y_6 \leq v$
- Probability Sum: $y_1 + y_2 + y_3 + y_4 + y_5 + y_6 = 1$
- Non-negativity: $y_j \geq 0$ for all $j = 1, \ldots, 6$; v is unrestricted.

These dual LP models are used to identify the optimal mixed strategy equilibrium, ensuring that no player can improve their outcome by unilaterally changing their probabilities.

B) Optimal Solutions and Interpretation

Optimal Payoffs (Value of the Game)

- According to the R results, the value of the game (v) for both players is 0
  - Player I's Optimal Payoff: 0
  - Player II's Optimal Payoff: 0 (calculated as -v in a zero-sum game)

- This indicates that the game is **fair**; when both players play optimally, neither has an inherent advantage.

Optimal Solutions (Mixed Strategies)

The optimal mixed strategies are vectors representing the probability with which each player should use their available strategies. Based on the R output variables:

- Player I's Optimal Strategy (x): (0.333, 0, 0, 0.333, 0.333, 0)
- Player II's Optimal Strategy (y): (0.333, 0, 0, 0.333, 0.333, 0)

Both players should utilize the same probability distribution:

- Strategy 1 (RWB): 33.3% of the time.
- Strategy 4 (WBR): 33.3% of the time.
- Strategy 5 (BRW): 33.3% of the time.
- Strategies 2, 3, and 6: 0% of the time.

Interpretation of Strategies

The spirit of game theory is to look at the worst outcomes and pick the "least worst" outcome to be as secure as possible.

- By mixing these three specific permutations (RWB, WBR, and BRW), each player ensures a security level of 0.

- This specific triplet of strategies functions like a "Rock-Paper-Scissors" cycle within the larger game. For example, RWB wins $ 160 against WBR but loses $ 160 against BRW.

- By playing each of these three strategies with equal probability (1/3), the players effectively neutralize any specific strategy the opponent might choose, ensuring they never lose more than $ 0 on average in the long run.

Uniqueness of the Solutions

The optimal solutions obtained are **not unique**.

- In this game, there is another independent triplet of strategies (RBW, WRB, and BWR) that also forms a balanced cycle.

- Running the solver might yield (0, 0.333, 0.333, 0, 0, 0.333) as an alternative optimal strategy.

- Since multiple "basic" optimal mixed strategies exist, any convex combination (weighted average) of these strategies is also an optimal solution. Therefore, there are infinitely many ways for the players to mix their strategies to achieve the same security level of 0.

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